

Technology Stack

Date	02 July 2026
Team ID	CCAP-01
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

The Credit Card Approval Prediction project is an end-to-end machine learning application that predicts whether a credit card applicant is likely to be Approved or Rejected. It covers data preprocessing, model training and comparison, and a Flask web application for single and batch predictions. The table below lists the technologies chosen for each layer of the architecture, along with the justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (HTML5, CSS3, Jinja2)	HTML5, CSS3 and Jinja2 templates rendered by Flask	The app uses server-rendered Flask/Jinja2 pages (home, login, signup, prediction, batch results, model info, features, admin) styled with a single custom stylesheet. This keeps the UI tightly integrated with the Flask backend without needing a separate build pipeline, which suits a data-science-focused project where the ML workflow is the core focus.
2	Backend / Server-Side (Python, Flask)	Python 3.11 with Flask 3.0 (served via Gunicorn)	Flask handles routing, session-based authentication, single and batch prediction endpoints, and history management. Python was chosen because the same language runs the entire ML pipeline (pandas, scikit-learn, XGBoost), avoiding any language hand-off between the model and the web

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			layer. Gunicorn serves the app in production as defined in the Procfile.
3	Database / Data Storage	SQLite (prediction_history.db) and CSV-based datasets	SQLite stores user accounts and prediction history without needing a separate database server, which is ideal for an application of this scale. The training data (application_record.csv and credit_record.csv) is merged on the ID column and used to derive the Approved / Rejected target label in preprocessing.py.
4	Cloud / Hosting / Deployment	Docker, Heroku-style Procfile and Gunicorn	A Dockerfile (python:3.11.9-slim base) containerizes the app for portable deployment on any cloud provider, while the Procfile and runtime.txt allow direct deployment to Heroku-style platforms. Gunicorn is used as the production WSGI server for stability under real traffic.
5	Version Control & CI/CD	Git & GitHub, GitHub Actions	GitHub hosts the source code and enables collaboration through commits and pull requests. A GitHub Actions workflow (.github/workflows/ci.yml) automatically runs the pytest test suite on each push, catching regressions in validation, prediction, and authentication logic early.
6	Third-Party APIs / Other Tools	Google OAuth 2.0, scikit-learn, XGBoost, SHAP, ReportLab, OpenPyXL	Google OAuth 2.0 enables secure social sign-in alongside standard email/password login. scikit-learn and XGBoost train and compare the Logistic Regression, Decision Tree, Random Forest, XGBoost, SVM, KNN, Naive Bayes, and Gradient Boosting models, while SHAP explains

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			individual predictions. ReportLab and OpenPyXL let users download their prediction results as PDF or Excel files.